

MAXIMUM PLAY · MINIMUM FOOTPRINT

AO Circular Design Spotlight on Supplies

A quick-reference guide on go-to suppliers

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Methods matter

Many materials currently used in the AO temporary build can be reused or recycled. It's how we treat, finish and handle them that turns resources into waste.

C MATERIALS CURRENTLY IN USE

The AO27 target material categories

KEEP USING THESE MATERIALS

- 1 Engineered wood sheets: plywood, formply, particleboard (E0)
- 2 Joinery/ cabinetry: formply, particleboard, melamine
- 3 Graphics: PVC sail track banners, removable SAV, ACP (if fire rated AS 1530.3-1999 & separated from substrate)
- 4 Surfaces: Textiles & fabrics
- 5 Reusable Flooring tiles: carpet & vinyl

HIGH-CIRCULARITY

- 6 Water-based paint
- 7 Steel: high value on the secondary market
- 8 Aluminium: high value on the secondary market

M METHODS

HOW we treat, finish & handle materials

Finishes decide the fate of a material. A sheet of plywood could be reused or is single-use bound for landfill depending on the method used to finish, fix or coat it.

METHODS THAT CREATE WASTE

- 01 Solvent-based adhesives – Cure into a permanent bond, so the substrate can't be cleanly separated for reuse
- 02 Solvent-based/alkyd paints penetrate and contaminate the timber fibre
- 03 Bonding composite materials
Two or more distinct materials bonded together create a waste problem if not separated

Composite materials

What is a composite?

Two or more distinct materials bonded together.
E.G. hi-res graphics surface rigidity (ACP+Ply) or
Façade finishes (stucco + particleboard) to create a
seamless look

Composites create a waste problem

- hard to separate cleanly
- recycling systems are 'mono-material'

The design challenge – what can you do

01 Design ACP to be separated from the substrate

Support the Panel Cycle take back program
! Be sure to only use AS 1530.1-1994 rated products
otherwise they will fail Fire/Safety testing

02 Innovation

Seek and assess hi-res display 'systems'

03 Experiment

Are there other facades or panels that can be used?
- Fabric, textured panel, water-based paint



Debonding ideas

Button-fix

Where to Buy

Products

Insights

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Our Products

The Button-fix concept is simple: robust Buttons are attached to the background structure and the mating Fixes are attached to the removable panel. Push on the panel firmly until the Button-fixes 'click'.

We offer a versatile [range of fasteners](#) to suit many different applications. We also offer complementary [buttons and accessories](#) to help make the Button-fix installation quick and easy.

8 products

Assembly	+
Attachment	+
Connection	+
Panel Thickness	+
Application	+



Type 1

Connects panels face-to-face with a 3mm or 8mm gap

Drop down or sideways engagement

Ideal for acoustic, wall and ceiling panels



Type 1 Bonded

For bonding to panels of any thickness

Drop down or sideways engagement

Ideal for signage and exhibition displays



Type 1 Flush

Connects panels face-to-face with no gap

Drop down or sideways engagement

Ideal for furniture assembly



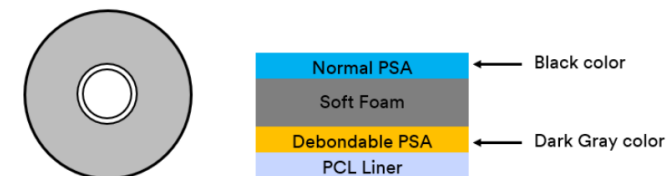
Type 1 Metal

For use on fire-retardant panels

Drop down or sideways engagement

Ideal for marine and other transport interiors

3M™ Heat Debondable Double Coated Tape 941XXHD



Tesa Industrial tapes

tesa

Markets Applications Products Sustainability Press & Insights

Benefits Applications Products FAQ

Filter

clear all

by construction attributes

Total thickness

Min 1 mm Max 3000 mm

Type of adhesive

acrylic (0)

biomass-balanced tackified acrylic (1)

10 Products

Product	Total thickness	Type of adhesive
tesa® 63615 > 1500 µm double sided PE foam tape (1.5mm thick)	1,500 µm	tackified acrylic
tesa® 62508 > Double sided PE foam mounting tape	800 µm	tackified acrylic
tesa® 51865 Next Gen - Team 4965 Differential >	165 µm	tackified acrylic

Guidance on finishes

Consider treatments, finishes and fixes that don't materials into waste.

DESIGN DECISIONS

RECOMMENDATION

 Oil/ solvent based paints	> Use water-based paints Won't contaminate materials
 Aluminium Composite Panel (ACP/ ACM) bonded to structural wood sheets	> Design to separate ACP from ply Aluminium has high value – use Panel-cycle recycling
 Self-adhesive vinyl (SAV)	> Design to peel off SAV from ply/ particleboard Aluminium has high value use Panel-cycle recycling
 Glues (solvents, formaldehyde additives)	> Mechanical connectors > Water-based, low VOC > Bio/ plant based Won't contaminate materials and enable modular assets
 Stucco/ Renders	> Removable SAV or sail-track PVC banners > Fabric > Moulded panels, textured cladding Use stucco and renders which have single-use in moderation

Graphics + surfaces

Hi-res graphics are a big part of the AO sharp reputation. There are few alternatives for PVC and vinyl however we can do these things to reduce waste:

HOW TO REDUCE WASTE

- 1 Use adhesives that allow SAV to be removed and materials to be recycled
- 2 Water-based paints won't contaminate materials
- 3 Renders and stucco will turn many recoverable materials into waste – use in moderation
- 4 Consider alternatives like fabrics

WHY USE THESE

IN USE TODAY

PVC banners (sail track)

Optima, B/Out
Edgeflex
Viewflex
RickyFlex

[Spicers Paper](#)
[Shann Australia](#)
[Halifax Vogel](#)
[Ricky Richards](#)

✓ Recovered & can be converted into paneling via ClimateOne

New

SAV - PVC-free self-adhesive

Removable & recyclable

[Visionary Digital](#)
[Avery](#)

✓ Can be removed and recycled into panels
✓ Less toxic production process

IN USE TODAY

Water-based paint

Preserves materials

[Wattyl ID](#)
[FX Australia](#)
[Dulux Enviro](#)

✓ Won't contaminate materials for future reuse
✓ Recyclable

IN USE TODAY

Paintback - recycled paint

Tinted & greyscale, made from AO's own recycled paint

[Paintback](#)

✓ 30% cheaper than commercial paint
✓ Won't contaminate materials
✓ Recyclable

IN USE TODAY

Fabric façades

Easy to wrap, suitable to many applications

[Tetrik Structures](#)

✓ Cheaper than SAV + substrates or ACP
✓ Great reusability
✓ Recyclable

Engineered wood

Many of the wood/ timber/ engineered wood materials in use today are reusable/ recoverable/ recyclable. Methods matter.

HOW TO REDUCE WASTE

- 1 Limit MDF where possible – evaluate these alternatives which have competitive pricing
- 2 Use water-based paint and consider finishes (see previous page)

WHY USE THESE

IN USE TODAY

Existing alternatives to MDF

Melamine particleboard (white -Chipboard substrate PF)
FormPly (particleboard substrate F14 -F17)

[Big River Group](#)
[Auswood](#)
[Evolution FP](#)

- ✓ Suppliers offer volume discounts
- ✓ These materials are already widely used – consider finishes that won't turn these recyclable materials into waste

Laminex Decorated Particleboard (Low Pressure Melamine)

[Laminex](#)

Durra panel – Ceilings, wall panels, acoustics

[Durra Panel](#)

New

New alternatives to MDF

SELEX Plywood – Structural B/D
EGGER OSB 3 E0 – Oriented Strand Board
EGGER Eurodekor Feelwood MFC Panels – Super E0
EGGER PerfectSense Lacquered ParticleBoard

[Forest One](#)

- ✓ Supplier offering volume discounts
- ✓ Potential alternatives to MDF
- ✓ Recyclable materials

IN USE TODAY

Cladding & facades

Weathertex – weatherboards & cladding

[Weathertex](#)

- ✓ Supplier offering volume discounts
- ✓ Alternative to fibre cement
- ✓ Made in Australia from sustainable materials
- ✓ Recyclable

Flooring

These flooring materials in use today are available at market rates – all with take-back or rehoming programs that keep materials out of landfill

HOW TO REDUCE WASTE

- 1 Use innovative (new) adhesives which support storage & reuse
- 2 Prefer ranges with a formal take-back or rehoming program

WHY USE THESE

IN USE TODAY

Milliken

Amplifier & Distinct Forms (tiles)
Major Frequency & Rigid Core LVT (planks)
+ more

[Citrus Commercial](#)
[JJ Contract Carpets](#)

- ✓ N/XT Closed Loop take-back recycled at no cost to TA
- ✓ Competitive pricing

IN USE TODAY

Tarkett

Desso Desert & Stratos EcoBase (tiles)
iD Square Loose-Lay LVT (planks)
Granit Safe.T (roll)
+ more

[Tarkett](#)

- ✓ No adhesives – suited to reuse
- ✓ ReStart take-back program
- ✓ Competitive pricing

IN USE TODAY

GH Commercial

Thematic Thread Carpet (tiles & planks)
Corsica Oak Engineered Timber Flooring
+ more

[GH Commercial](#)

- ✓ Rehome program – donates to social enterprises
- ✓ Competitive pricing

IN USE TODAY

Astro Turf

Outdoor synthetic turf

[Tuff Group](#)

- ✓ Take-back and recycling program
- ✓ Competitive pricing



What are your spotlight materials?

Have a material you'd like reviewed for AO27? Tell us about it

Circular Design Guidelines

Access https://megtei.github.io/Materials_Methods_Plan